

An Improved Approximation Algorithm for the Metric Uncapacitated Facility Location Problem

Maxim Sviridenko

IBM T. J. Watson Research Center, Yorktown Heights,
P.O. Box 218, NY 10598, USA
sviri@us.ibm.com

<http://www.research.ibm.com/people/s/sviri/sviridenko.html>

Abstract. We design a new approximation algorithm for the metric uncapacitated facility location problem. This algorithm is of LP rounding type and is based on a rounding technique developed in [5,6,7].

1 Introduction

In the uncapacitated facility location problem we are given a set of potential facility locations $\mathcal{F}$ and a set of demand points $\mathcal{D}$. Each point $i \in \mathcal{F}$ has an associated opening cost f_i . Once facility point i has been opened it can provide an unlimited amount of certain commodity. A client $j \in \mathcal{D}$ has a demand of commodity that must be shipped from one of the opened facilities. We assume that transportation costs of demand transferred from facility point i to demand point j are proportional to the distance c_{ij} between them. We assume that c is a metric, i.e. $c_{ii} = 0$ for all $i \in \mathcal{F}$, $c_{ij} = c_{ji}$ for all $i, j \in \mathcal{D} \cup \mathcal{F}$ and $c_{ij} + c_{jk} \geq c_{ik}$ for all $i, j, k \in \mathcal{D} \cup \mathcal{F}$. The goal is to determine a subset of potential facility points that minimizes the sum of total opening costs of facilities at these locations and total service costs of all demand points from opened facilities.

This problem can be written as the following mixed integer program

$$\min \sum_{i \in \mathcal{F}} \sum_{j \in \mathcal{D}} c_{ij} x_{ij} + \sum_{i \in \mathcal{F}} f_i y_i, \quad (1)$$

$$\sum_{i \in \mathcal{F}} x_{ij} = 1, \quad j \in \mathcal{D}, \quad (2)$$

$$x_{ij} \leq y_i, \quad i \in \mathcal{F}, j \in \mathcal{D}, \quad (3)$$

$$x_{ij} \geq 0, \quad i \in \mathcal{F}, j \in \mathcal{D}, \quad (4)$$

$$y_i \in \{0, 1\}, \quad i \in \mathcal{F}. \quad (5)$$

The uncapacitated facility location problem (UFLP) (another popular name is the simple plant location problem) is one of the most studied and known optimization problems. There are hundreds of papers on different aspects of this problem (see the survey chapter of Cornuejols, Nemhauser & Wolsey in [31] and short survey paper on polynomially solvable subcases of the problem by Ageev & Beresnev [3]). In this paper we are mainly interested in approximation

algorithms with proven performance guarantees. We say that an algorithm has performance guarantee $\rho \geq 1$ if it always delivers a solution within a factor of ρ of optimum. We also call such an algorithm as ρ -approximation algorithm.

The first approximation algorithm for UFLP was built by Cornuejols, Fisher & Nemhauser [16]. They obtain $(1 - e^{-1})$ -approximation algorithm for the maximization variant of UFLP. Recently, Ageev & Sviridenko [4] obtained an 0.828-approximation algorithm for this problem. The first approximation algorithm for the minimization version of the general UFLP (c might not satisfy the triangle inequality in this case) was built by Hochbaum [21]. Her algorithm has performance guarantee at most $\ln n + 1$. Slightly better algorithm was recently discovered by Young [33]. And it is easy to prove by a reduction from the set cover problem that the general UFLP is at least as hard to approximate as the set cover problem. On the other side Feige [17] proved that for any $\varepsilon > 0$ there is no $(1 - \varepsilon) \ln n$ -approximation algorithm for the set cover problem unless $NP \not\subseteq DTIME(n^{\log \log n})$.

The most interesting special case of the general UFLP is *the metric UFLP*. In this problem distance function c satisfies the triangle inequality. The first approximation algorithm for this problem was obtained by Shmoys, Tardos & Aardal [32] using clustering and filtering techniques due to Lin & Vitter [26]. The performance guarantee of their algorithm is 3.16. Guha & Khuller [18] obtain 2.408-approximation algorithm by combining Shmoys, Tardos & Aardal's algorithm with the greedy algorithm. They also proved that there is no an approximation algorithm with performance guarantee better than 1.463 unless $NP \not\subseteq DTIME(n^{\log \log n})$ by the reduction from the set cover problem. Chudak & Shmoys [12,14] improved the Shmoys, Tardos & Aardal's algorithm by using dependent randomized rounding and information about the dual linear program to the linear programming relaxation of the problem (1)-(5). Their original performance guarantee [12] was $1 + 2e^{-1} \approx 1.736$ but using some additional tricks they slightly improved it to 1.732 [14]. Charikar & Guha [10] obtained 1.728-approximation algorithm by combining the primal-dual algorithm, the greedy algorithm and the Chudak & Shmoys's algorithm. Jain, Mahdian and Saberi [22] obtained 1.61-approximation algorithm using technique developed by Mahdian et al. [27]. Recently, combining few different approaches Mahdian, Ye & Zhang [28] obtained 1.52-approximation algorithm.

There are also few very interesting approximation algorithm for the metric UFLP based on different ideas with worse performance guarantees: local search algorithms was analyzed by Korupolu et al. [25], Arya et al. [9] and Charikar & Guha [10], primal-dual algorithms was studied by Jain & Vazirani [23] and Charikar & Guha [10]. Mettu & Plaxton [29] analyzed an algorithm very similar with Jain & Vazirani's one but their analysis doesn't use the primal-dual technique. The advantage of such algorithms is that all of them are combinatorial and very fast. All these techniques, developed originally for UFLP, was applied to different generalizations of this problem like capacitated facility location problem, multilevel facility location problem and others [32,23,15,13,1,25,30].